

Instructor: Joshua Flynn
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Office hours: TBA
Course webpage: <https://jlflynn.com/18.100a/>
Lectures: Tuesdays and Thursdays, 1:00–2:30 PM, 24-121

Visit the course webpage for more information about the course and its policies.

Course materials

The main source for the course is Casey Rodriguez's Fall 2020 MIT 18.100A OpenCourseWare materials: <https://ocw.mit.edu/courses/18-100a-real-analysis-fall-2020/>.

A supplemental reference is Jiri Lebl's *Basic Analysis: Introduction to Real Analysis*: <https://www.jirka.org/ra/>.

Grading

Component	Weight
Psets	12.5%
Quizzes	12.5%
Exam 1 (<i>Thu Oct 8</i>)	25%
Exam 2 (<i>Thu Nov 12</i>)	25%
Exam 3 (<i>administered during the final-exam period</i>)	25%

When computing the final course grade, I will drop either your lowest pset grade or your lowest quiz grade, whichever is more favorable. I will not drop both.

Psets

Psets are assigned weekly and are due Tuesdays. The first pset is due September 22.

Each pset contains five graded problems and three ungraded problems. Of the five graded problems, two are randomly selected for grading for correctness and three are graded for completion.

All five graded problems are relevant to quizzes. Both graded and ungraded problems may be relevant to exams.

Collaboration: Collaboration with other students in the class is encouraged. However, each student must write up and submit their own solutions, and all collaborators should be documented on the submitted work.

Submission: Students must submit scanned copies of **handwritten** solutions to Gradescope, accessible via Canvas. Typed solutions will not be accepted.

Quizzes

Quizzes are administered weekly on Tuesdays and last 15 minutes. The first quiz will be administered September 22.

Each quiz consists of two problems that will be, possibly with variations, two of the three pset problems graded for completion.

Exams

All exams are in person and closed book.

Exams 1, 2, and 3 each consist of four problems and have the same general problem-type structure: three problems based on problem types practiced on psets or quizzes, and one novel problem requiring students to apply course ideas in an unfamiliar setting.

Exams 1 and 2 are 80 minutes long and are administered during regular class time. Exam 3 is administered during the final-exam period. Exam 2 covers material taught after Exam 1, and Exam 3 covers material taught after Exam 2. Note, however, that material in this course builds on itself. You may therefore be expected to use techniques developed earlier in the course, but the exams are not cumulative in the traditional sense.

Reassessment

During the final-exam period, students may optionally complete reassessments for Exam 1 and/or Exam 2. Students may attempt either reassessment, both, or neither.

A reassessment can never lower the original exam grade. A reassessment may raise the recorded exam grade to at most 77%.

The recorded grade for a reassessed exam is

$$\max(\text{original grade}, \min(\text{reassessment grade}, 77)).$$

The final-exam period is three hours. Students may allocate this time between Exam 3 and any reassessments they choose to attempt. Students who attempt neither reassessment may use the full three hours on Exam 3. In particular, students who scored 77% or higher on both Exams 1 and 2 cannot improve those grades through reassessment and may use the full three hours on Exam 3.

Late work and missed quizzes/exams

Late psets are not accepted. Gradescope provides a 30-minute late-submission window, but any submission made after the stated deadline will be marked as late.

Rescheduling a missed quiz or exam requires support from Student Support Services (S3) documenting a compelling reason, such as illness or another circumstance outside the student's control. Athletic or professional obligations may be accommodated when coordinated in advance with the appropriate MIT staff. Holiday travel, job interviews, and conflicts resulting from registration in two simultaneous classes are not grounds for rescheduling a quiz or exam.

Math help

Students may seek help through office hours or scheduled appointments. The MIT Math Learning Center is also available as a resource.

Student Support Services

If you are dealing with a personal or medical issue that is impacting your ability to attend class, complete work, or take a quiz or exam, please discuss this with Student Support Services (S3). The deans in S3 can verify your situation and discuss with you how to address missed work. You may consult with S3 in 5-104 or at 617-253-4861. S3 offers scheduled appointments and virtual drop-ins; see the S3 webpage for current availability.

Disability and Access Services

MIT is committed to the principle of equal access. Students who need disability accommodations are encouraged to speak with Kathleen Monagle, Associate Dean and Director of Disability and Access Services, prior to or early in the semester so that accommodation requests can be evaluated and addressed in a timely fashion. Even if you are not planning to use accommodations, it is recommended that you meet with DAS staff to familiarize yourself with the services and resources of the office. You may contact DAS at das-student@mit.edu or 617-253-1674. If you have a disability accommodation letter from DAS, please speak with Theresa Cummings (tcumming@mit.edu) in Math's Academic Services office as soon as possible to make arrangements for the semester.

AI use

AI use is **not permitted on exams or quizzes**.

AI use is **permitted on psets**. Students who use AI should make a serious effort to understand the mathematics rather than simply obtaining or copying solutions. AI-generated explanations can also create a false sense of understanding, so students are encouraged to use AI carefully and critically.